



ABDULLAH RACHMAT SUGI

GEOMATICS ENGINEERING | GIS | SURVEY & MAPPING

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SUMMARY

Fresh Graduate in Geomatics Engineering from Institut Teknologi Sepuluh Nopember (ITS) with a GPA of 3.41/4.00 and a Very Satisfactory distinction, with experience in GIS, surveying and mapping, and geospatial data management and analysis. Experienced through internships at ATR/BPN and PT Kereta Api Indonesia in GNSS surveying, land asset mapping and verification, cadastral data validation, and spatial data processing using ArcGIS and AutoCAD. Additional project experience includes remote sensing, UAV photogrammetry, and 3D modeling, with the ability to process, analyze, and validate geospatial data to produce accurate spatial information and support mapping and analytical needs.

EDUCATION

Sepuluh Nopember Institute of Technology

Jul 2022 - Aug 2026

Bachelor of Engineering in Geomatics Engineering

- GPA: 3.41/4.00
- Thesis on "3D Modeling of Candi Tikus Using Integrated UAV and Terrestrial Photogrammetry Data Based on the Gaussian Splatting Method"

SMA Negeri 1 Bangsal

Jul 2019 - May 2026

Science Major (MIPA)

- Average Grade : 90.23/100

INTERNSHIP EXPERIENCE

PT Kereta Api Indonesia (Persero) Regional Operation VII Madiun

Jul 2025 - Aug 2025

- Conducted terrestrial boundary surveys for PT KAI land assets on approximately 30 parcels in the Jombang Station area and Jl. Anggrek Madiun using EFIX GNSS RTK for field boundary verification.
- Processed survey data and spatial digitization using AutoCAD Civil 3D and ArcGIS Pro to delineate asset boundaries, integrate administrative data, and analyze corridor areas for operational and non-operational lines.
- Developed 6 thematic maps and 1 survey layout depicting station distributions, land asset boundaries, and railway corridors across Daop 7 Madiun.

National Land Agency (ATR/BPN) of Kediri City

Jan 2025 - Feb 2025

- Processed and updated cadastral maps using AutoCAD and ArcGIS to support land registration data maintenance across Kediri City.
- Conducted terrestrial field boundary surveys using GNSS for residential housing areas and built structures in Kediri City for on-site boundary verification.
- Cross-checked spatial parcel data against land certificates (SHM) to ensure consistency between physical boundaries and legal ownership documents.

National Land Agency (ATR/BPN) of Batu City

Jul 2024 - Aug 2024

- Performed topological validation and data cleaning on 115 land parcels to eliminate gaps and overlaps, ensuring spatial accuracy and database consistency.
- Conducted field boundary surveys using geodetic GNSS equipment across 10 plantation parcels in Sumberbrantas Village, Bumiaji District.
- Validated land ownership records by cross-referencing freehold certificates (Sertifikat Hak Milik/SHM) with the institutional cadastral database to ensure legal tenure and data integrity.

ORGANIZATIONAL EXPERIENCE

Student Legislative Board, Faculty of Civil, Planning, and Geo Engineering (BLM FT-SPK ITS)	Nov 2024 - Feb 2026
- Responsible for developing the Organizational Work Guidelines (GBHK) for HIMAGE-ITS and providing guidance to commission members. - Conducted policy and regulatory reviews, drafting, and evaluation to support effective organizational governance.	
Student Representative Council, Geomatics Engineering Student Association (BPM HIMAGE-ITS)	Dec 2023 - Dec 2025
- Served as Session Chair II and contributed to the implementation of the High-Level Deliberation (MTT) of FT-SPK ITS. - Managed official correspondence and administrative processes while establishing strategic communication with organizational stakeholders.	

CERTIFICATIONS & TRAINING

Remote Pilot Certificate (RPC) – Small Unmanned Aircraft System
Directorate General of Civil Aviation | 2026 (Valid thru 2027)

Remote Pilot Rating – Small Unmanned Aircraft System
PT Terra Drone Indonesia | 2026 (Valid thru 2028)

Hyperspectral Data for Land and Coastal Systems
NASA ARSET (Applied Remote Sensing Training) | 2026

PROJECTS

3D Modeling of Candi Tikus Using Integrated UAV and Terrestrial Photogrammetry Data Based on the Gaussian Splatting Method	Nov 2025 - June 2026
Bachelor's Thesis Project - Integrated 1,543 UAV and terrestrial photogrammetric images using 3D Gaussian Splatting (3DGS) with 20,000 optimization iterations. - Produced a model 70% smaller with 4–6× faster rendering compared to SfM/MVS. - Achieved 1.36 cm RMSE and 0.183 m Cloud-to-Cloud deviation, validated across 7 field measurement segments.	
Spatio-Temporal Monitoring of Wonorejo Aquaculture Ponds	Feb 2025 - May 2025
2025 Thematic Fieldwork – Department of Geomatics Engineering, ITS - Processed multitemporal Sentinel-2A imagery (2015–2024) using SVM in ENVI for land-cover classification. - Executed confusion matrix-based accuracy assessment, achieving an Overall Accuracy of up to 81.26% and a Kappa coefficient of 0.66. - Produced cartographically standardized multitemporal thematic maps using ArcGIS Pro.	
Complete City Project (Kota Lengkap)	Mar 2024 - Apr 2024
Collaboration Project with BPN Surabaya II and Department of Geomatics Engineering, ITS - Conducted cadastral data cleaning and validation in Gading Village, Surabaya. - Identified 121 topological anomalies (104 overlaps and 17 gaps) and performed parcel geometry corrections. - Synchronized ArcGIS shapefiles with AutoCAD working drawings.	

ADDITIONAL INFORMATION

- Technical Skills:** ArcGIS Pro, QGIS, ENVI, Agisoft Metashape, Autodesk Civil 3D, Google Earth Pro, Leica Infinity, HYPACK, Surfer, MATLAB, Python (Basic), Visual Studio Code, HTML, CSS, JavaScript, Microsoft Office, and Vercel Deployment.
- Languages:** Bahasa Indonesia (Native), English (Professional Working Proficiency)